

Advanced (Carolina Reaper)

Q1. Solve for x in the equation $2^x = 32$.

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

(A)

1102.50

(B)

1125

(C)

1150

(D)

1175

(E)

1200

Q2. If $3^x = 81$, what is the value of x ?

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q3. Solve for y in the equation $4^y = 256$.

- (A) $y = 2$
- (B) $y = 3$
- (C) $y = 4$
- (D) $y = 5$
- (E) $y = 6$

Q6. A population of bacteria is growing exponentially. If the population grows from 100 to 400 in 2 hours, what is the hourly growth rate of the population?

- (A) 50%
- (B) 75%
- (C) 100%
- (D) 125%
- (E) 150%

Q4. A store is having a sale where the price of an item is reduced by 25% each day. If the original price of the item was

80

, what is the price of the item after 2 days?

(A)

40

(B)

45

(C)

50

(D)

55

(E)

60

Q5. A person invests

1000

in a savings account that earns an annual interest rate of 5%. How much will the person have in the account after 2 years, assuming compound interest?

Q7. Solve for z in the equation $5^z = 125$.

- (A) $z = 2$
- (B) $z = 3$
- (C) $z = 4$
- (D) $z = 5$
- (E) $z = 6$

Q8. A company is expanding its operations and expects its revenue to grow exponentially. If the company's revenue grows from

100,000

to

400,000

in 2 years, what is the annual growth rate of the revenue?

- (A) 50%
- (B) 75%
- (C) 100%
- (D) 125%
- (E) 150%

Open Ended Questions (FRQ)

Q9. A person is planning a road trip and wants to calculate the total cost of the trip. The cost of gas is

2.50

per gallon, and the person's car gets an average of 25 miles per gallon. If the person drives a total of 500 miles, what is the total cost of gas for the trip? Show your work and explain your reasoning.

Q10. A store is having a sale where the price of an item is reduced by 20% each day. If the original price of the item was

120

, what is the price of the item after 3 days? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Remind students to use the properties of exponents to simplify equations.
- Tip 2: Encourage students to use real-world examples to illustrate the concept of exponential growth.
- Tip 3: Emphasize the importance of showing work and explaining reasoning in free-response questions.